CO3133 / ASSIGNMENT 02

Fine-Grained Food-101 Classification

A from-scratch CNN baseline vs. a fine-tuned ImageNet ResNet-50

Group G-03 | Semester 261

Nguyen Van An • Vo Le Hai Dang • Huynh Vuong Khang

Ho Chi Minh City University of Technology (HCMUT), VNU-HCM

Instructor: Lê Thành Sách | November 2026

Problem: 101 dishes, many visually similar

Task formulation

Input

One RGB photograph of a single dish

128×128 (CNN) or 224×224 (ResNet-50)

Output

One of 101 Food-101 dish categories

Prediction = argmax of 101 logits

Objective

Minimise categorical cross-entropy

Evaluate top-1 accuracy + macro-F1

Why is it difficult?

- **Fine-grained similarity**
  Steak, filet_mignon and prime_rib can share colour, cut and plating.
- **Within-class variation**
  Lighting, camera angle, garnish and presentation vary widely.
- **Background clutter**
  Tableware and surroundings may occupy more pixels than the dish.
- **Label noise**
  Official training labels are automatically assigned and intentionally uncleaned.

Food-101: scale, balance and visual variability

IMAGES
101,000
101 dish categories

OFFICIAL TRAIN
75,750
750 per class

OFFICIAL TEST
25,250
250 per class

STORAGE
~10 GB
JPEG / 11 parquet shards

Dataset properties

- ETH Zürich / Foodspotting
- Original 2014 release
- One dish label per image; no boxes or masks
- Balanced by construction: 750 train + 250 test per class
- Training labels are noisy; test labels manually verified

EDA observations

- Variable image resolution and aspect ratio (max side 512 px)
- Warm hues dominate; broad saturation/value range
- Background clutter is common
- After pHash analysis: 271 candidate within-class near-duplicate pairs
- 102 train–test candidate pairs (≤0.40% of test)

Split and preprocessing: protect the final test set

Training protocol split



Validation = stratified 10% carve from official train (seed 42). Test split retained unchanged and evaluated once per run.

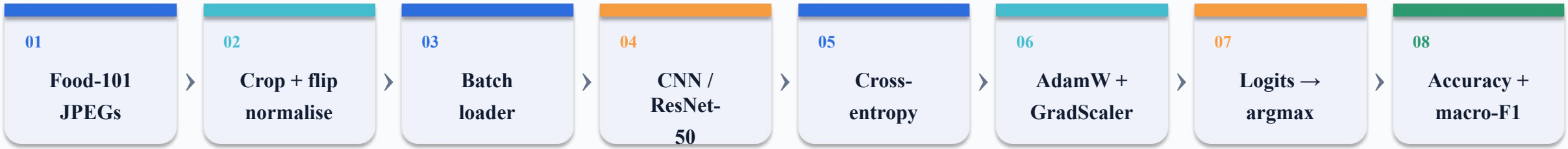
Training-time transforms

RandomResizedCrop (scale 0.7–1.0)
RandomHorizontalFlip
ImageNet mean / standard deviation normalisation

Evaluation transforms

Resize to round(size × 256 / 224)
CentreCrop to target size
Same evaluation policy across configurations

End-to-end classification pipeline



Shared across all configurations

Same dataset split, loss, batch size, epoch budget, seeds and checkpoint rule.

What changes

Input size and model initialisation differ for the baseline; frozen vs. full ResNet-50 changes backbone trainability.

Best checkpoint: restore the epoch with highest validation macro-F1; test evaluation happens once after selection.

Three configurations, two different roles

A | From-scratch CNN

- Four Conv–ReLU–MaxPool blocks
- Channels: 32 → 64 → 128 → 256
- 128×128 input
- Flatten 256×8×8 → FC 512 → 101
- 8.83M parameters; all trainable

Purpose: reference performance without external pretraining.

B | ResNet-50 frozen

- ImageNet IMAGENET1K_V2 weights
- Replace 1,000-way head with 101-way head
- 224×224 input
- Freeze all backbone parameters
- Train 206,949-parameter head only
- BatchNorm held in eval mode

Purpose: linear probe on fixed ImageNet features.

C | ResNet-50 full

- Same pretrained ResNet-50 and new head
- 224×224 input
- Train all 23.715M parameters
- AdamW, global LR = 10^{-4}

Purpose: adapt the pretrained representation to Food-101.

Implementation: CNN and training/evaluation pipeline self-implemented; ResNet-50 backbone and pretrained weights from torchvision.

Controlled experiment: frozen vs. full fine-tuning

Hypothesis

Full fine-tuning will outperform a frozen backbone on test accuracy and macro-F1.

Changed factor

Backbone trainability: frozen vs. all parameters trainable.

Decision criterion

Compare 3-seed means; support if full exceeds frozen by more than across-seed SD.

Fixed factors in the ResNet-50 ablation

Factor	Both configurations
Data / split	Food-101; split seed 42
Input / augmentation	224×224; same crop, flip, normalisation
Optimisation	AdamW; LR 1e−4; weight decay 1e−4
Training budget	Batch 128; 30 epochs; seeds 42, 67, 69
Other	Cross-entropy; A100; mixed precision

Baseline is not an ablation

The scratch CNN also differs in pretraining, resolution (128 vs. 224), learning rate and parameter count. Use it as a reference, not as an isolated architecture comparison.

Training behaviour reveals different convergence patterns

Scratch CNN

Selected epoch: 26 / 25 / 30

Val accuracy peaks late; validation loss minimum occurs earlier (epochs 12–15).

Gap at epoch 30: 0.213–0.238

ResNet-50 frozen

Selected epoch: 30 / 30 / 30

Validation loss and accuracy still improve at the end of the 30-epoch budget.

Gap at epoch 30: 0.041–0.044

ResNet-50 full

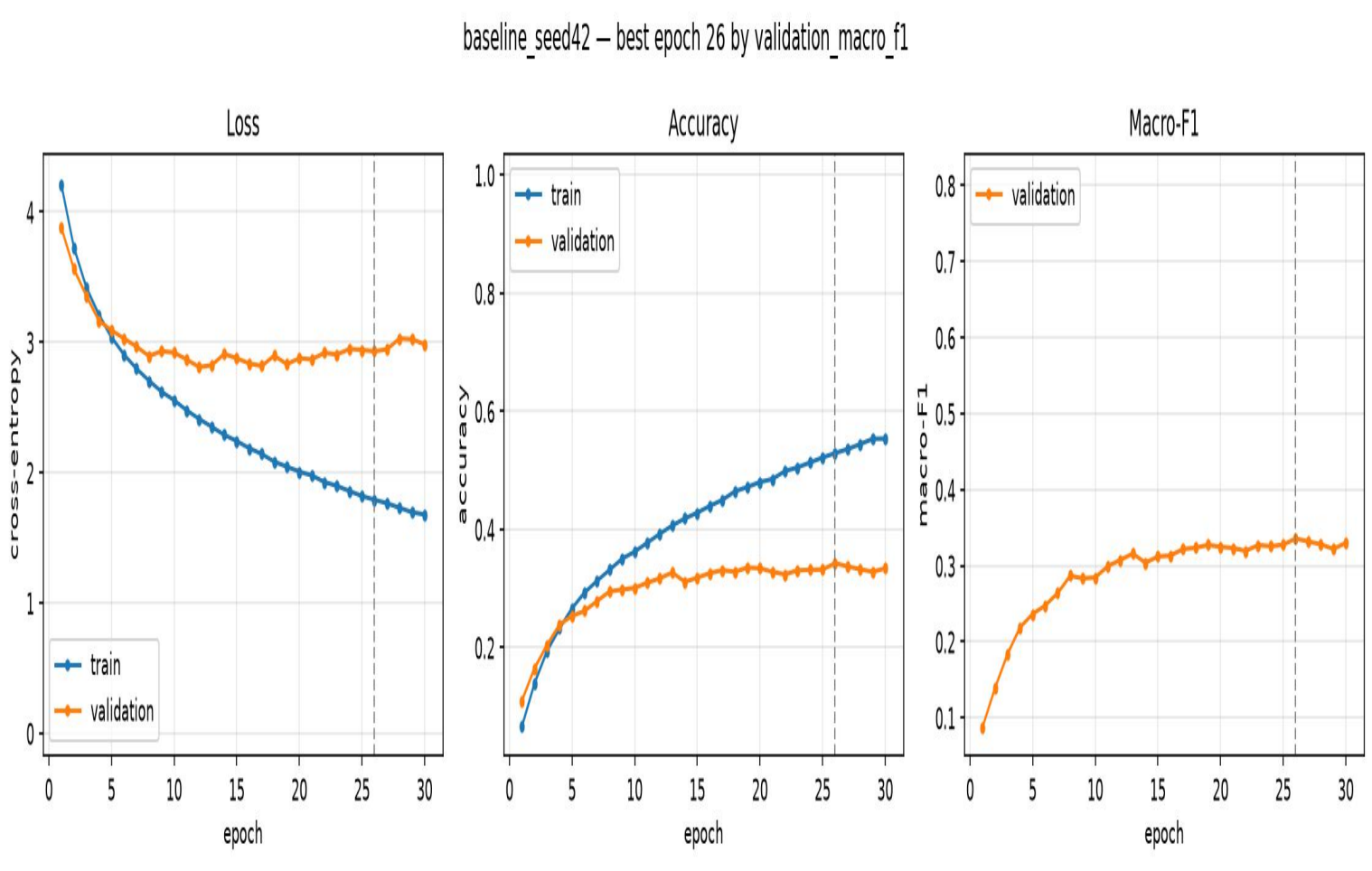
Selected epoch: 4 / 9 / 4

Train accuracy reaches ~98.8%; validation performance peaks early, then degrades.

Gap at epoch 30: 0.186–0.192

Interpretation Full fine-tuning has strong capacity but overfits after its early peak; the frozen model is likely under-trained within this fixed budget.

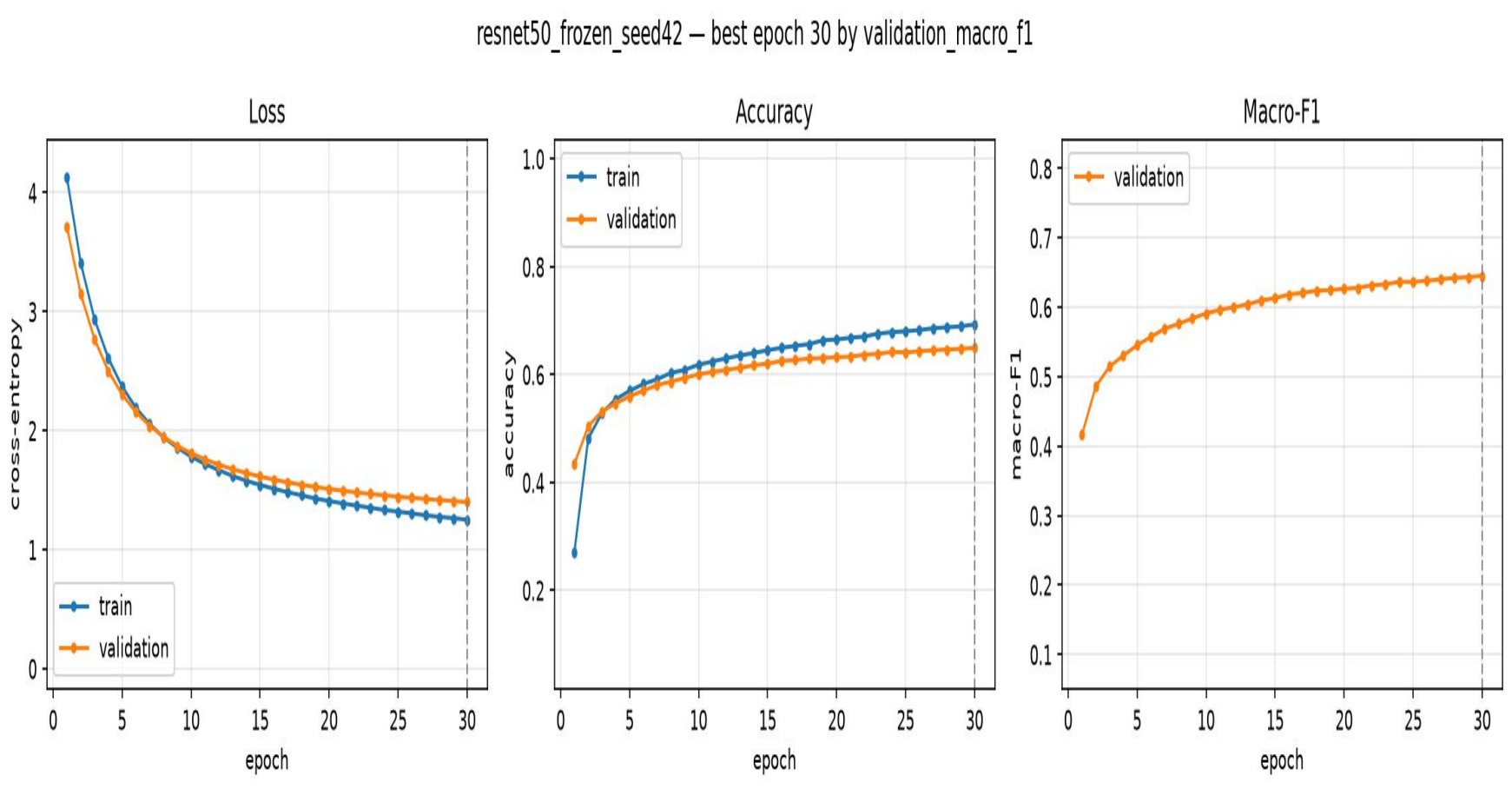
Baseline convergence



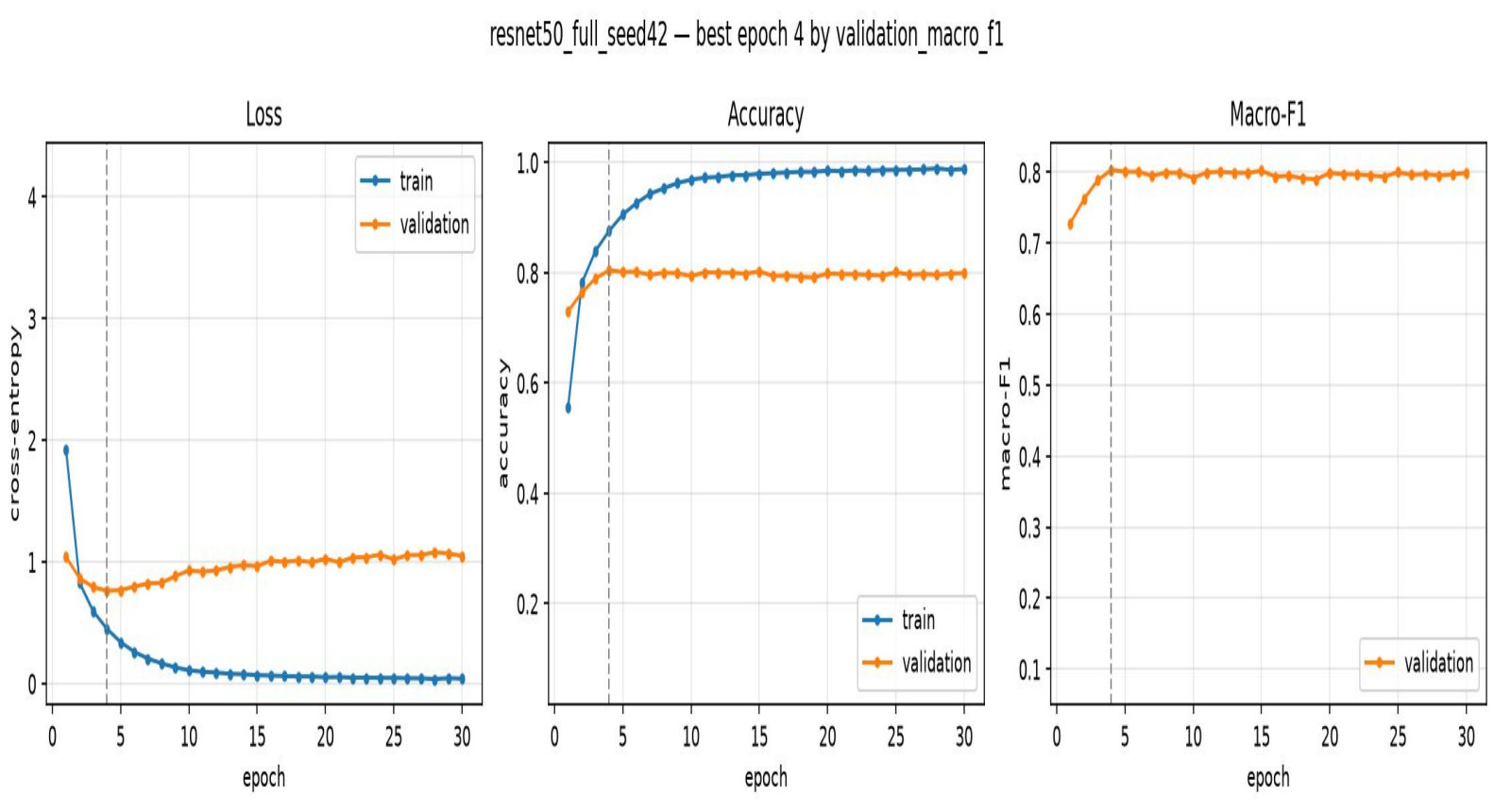
- **Peak Epoch:** Selected late at Epoch 25–30 (Epoch 26 on Seed 42).
- **Behavior:** Slow learning plateau with an early disconnect .
- **Train-Val Gap:** Widens to 0.213–0.238 by Epoch 30.

Convergence curves: ResNet-50 frozen backbone

- **Peak Epoch:** Reaches its best result at the final Epoch 30 across all 3 seeds.
- **Behavior:** Monotonic, steady improvement in both loss and accuracy throughout all 30 epochs with no overfitting.
- **Train-Val Gap:** Remains tiny at 0.041–0.044



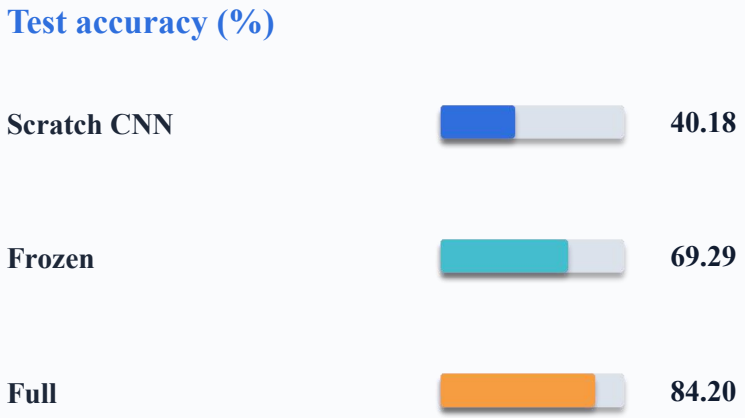
Convergence curves: ResNet-50 full fine-tuning



- **Peak Epoch:** Reaches best performance extremely fast at Epoch 4–9 (Epoch 4 on Seed 42).
- **Behavior:** Rapid initial convergence to ~80.4% validation accuracy, followed by severe late-epoch overfitting (train accuracy hits 98.8%).
- **Train-Val Gap:** Expands significantly to 0.186–0.192 at Epoch 30.

Main result: full fine-tuning leads on both primary metrics

Configuration	Test accuracy	Test macro-F1	Best epoch
Scratch CNN	40.18 ± 2.38%	39.74 ± 2.48%	27.0
ResNet-50 frozen	69.29 ± 0.06%	68.96 ± 0.08%	30.0
ResNet-50 full	84.20 ± 0.10%	84.19 ± 0.15%	5.7



FULL VS FROZEN

+14.91 pp

test accuracy

MACRO-F1 GAIN

+15.23 pp

test macro-F1

FULL SEED RANGE

84.10–84.28%

test accuracy

FROZEN SEED RANGE

69.25–69.35%

test accuracy

Accuracy and macro-F1 are close because Food-101 is balanced; macro-F1 confirms the result is not driven by a few large classes.

Ablation: allowing the backbone to adapt changes the outcome

FROZEN TEST ACC.	FULL TEST ACC.	ACCURACY GAIN	TRAINING COST
69.29%	84.20%	+14.91 pp	+35.6%
206,949 trainable params	23,714,981 trainable params	same backbone + data	1,698 s vs. 1,252 s

What the ablation suggests

- **Representation adaptation**
  ImageNet features are useful, but adapting the backbone substantially improves this fine-grained task.
- **Not a clean maximum-capacity comparison**
  Frozen training is still improving at epoch 30; it has not converged under the shared budget.
- **Generalisation trade-off**
  Full fine-tuning shows a much larger train–validation gap at epoch 30.

Careful conclusion: the experiment supports full fine-tuning under this protocol; it does not establish the best achievable frozen-backbone result.

Compute cost and practical trade-offs

Model	Total params	Trainable params	Train time (s)	Inference (ms/img)
Scratch CNN	8.83M	8.83M	1,342.5	0.416
ResNet-50 frozen	23.71M	0.207M	1,252.0	0.730
ResNet-50 full	23.71M	23.71M	1,698.1	0.623

Lowest training time

Frozen ResNet-50

1,252 s for 30 epochs

Highest accuracy

Full ResNet-50

84.20% test accuracy

Fastest measured inference

Scratch CNN

0.416 ms/image

Error analysis: the remaining failures are structured

FULL ERRORS

4,016

seed 42 / 25,250

FROZEN ERRORS

7,738

seed 42 / 25,250

BASELINE ERRORS

15,768

seed 42 / 25,250

TOP 20 PAIRS

10.8%

432 / 4,016 full errors

Frequent confusion families in full fine-tuning

Red-meat cuts

filet_mignon ↔ steak

prime_rib → steak

pork_chop → steak

Similar cuts, garnish and plating.

Raw preparations

tuna_tartare ↔ beef_tartare

Similar moulded-disc presentation and colour.

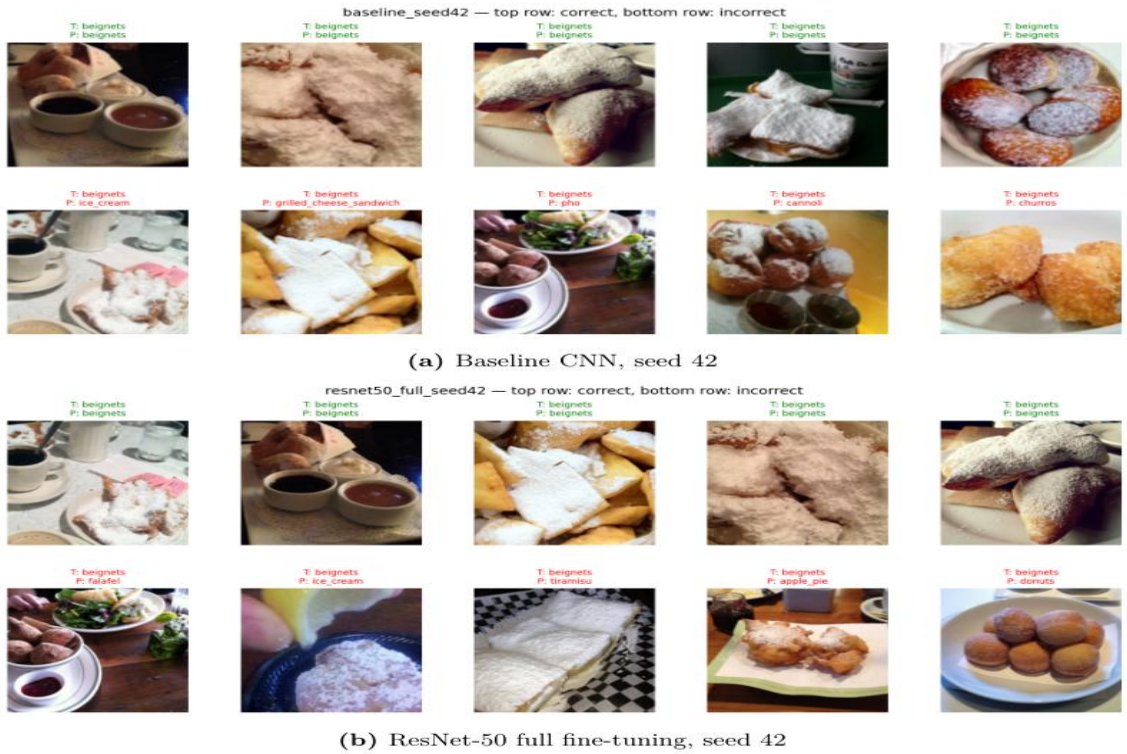
Desserts / fried foods

chocolate_cake → chocolate_mousse

Other visually similar plated desserts and battered items.

Baseline additionally makes coarse cross-category errors (e.g., pad_thai → fried_rice); ResNet-50 errors are more concentrated among visually related dishes.

Qualitative predictions: examples from the test batch



(a) Baseline CNN, seed 42

(b) ResNet-50 full fine-tuning, seed 42

Figure 3: Test predictions versus ground truth for seed 42. Top row of each panel: five correct predictions. Bottom row: five incorrect predictions. Each title gives the true label (T) and the predicted label (P).

What this figure shows

For seed 42: five correct (top row) and five incorrect (bottom row) examples.

How to interpret it

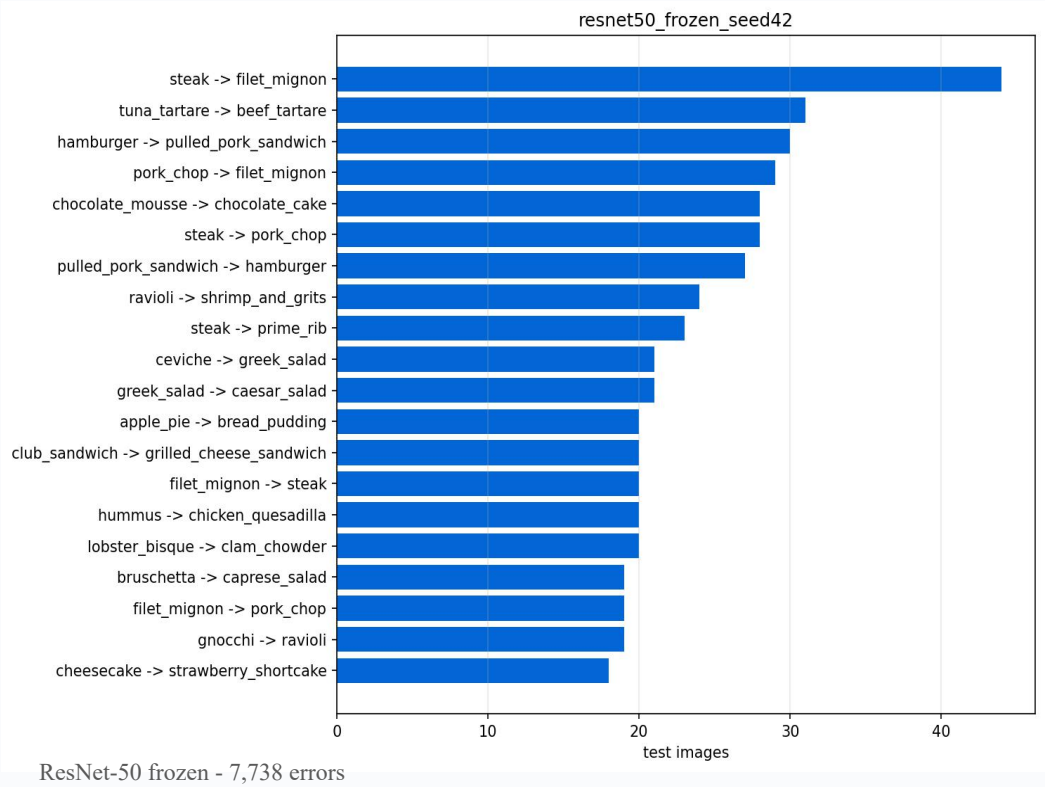
These are illustrative samples from the first test batch, not a difficulty-ranked sample or full-test summary.

Whole-test evidence

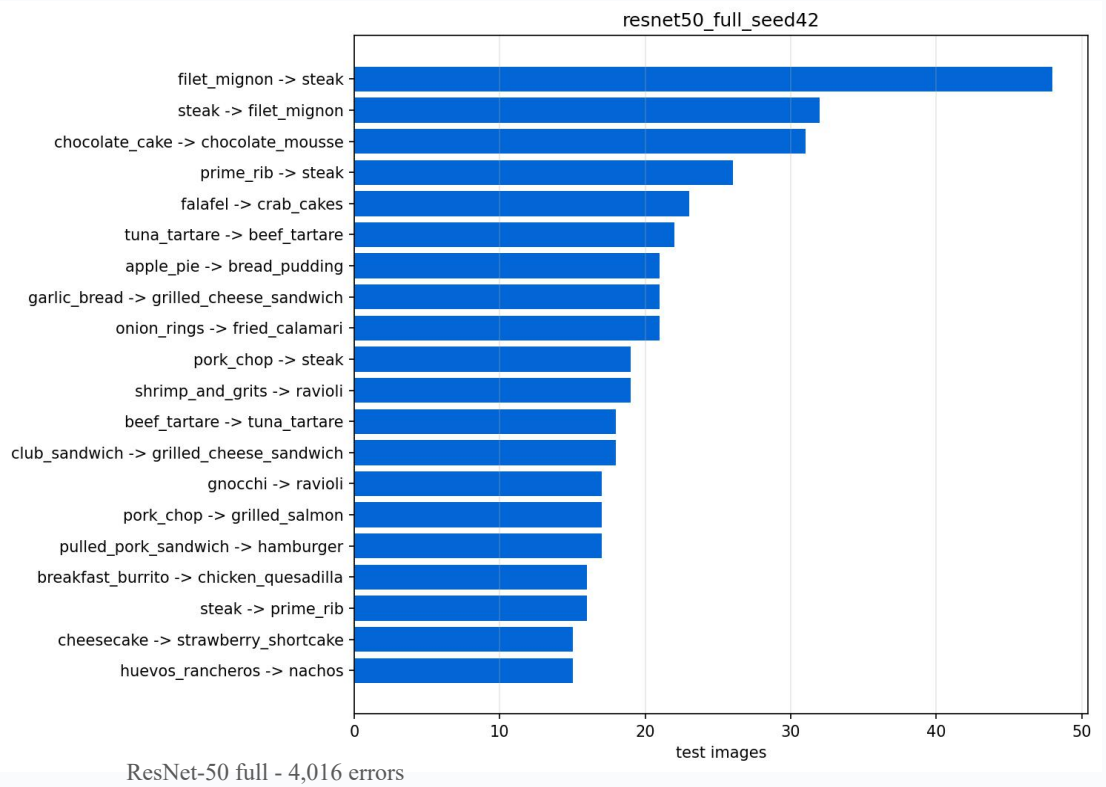
Per-class metrics and confusion analysis cover all 25,250 test images.

Do not overgeneralise from the displayed examples; use the aggregate metrics and confusion pairs for conclusions.

Where do the models confuse classes?



ResNet-50 frozen - 7,738 errors



ResNet-50 full - 4,016 errors

- Shared families: **Frozen and full ResNet-50 share red-meat and raw-preparation confusions.**
- Full fine-tuning lower the error counts overall and for several prominent confusions (steak -> filet_mignon 44 -> 32; tuna_tartare -> beef_tartare 31 -> 22) without changing which classes are confused.
- **Most errors are diffuse:** The top 20 pairs are only 10.8 % of the errors of full fine-tuning; the remaining 89.2 % lie elsewhere in the 101 x 101 matrix.

Conclusion: strong evidence, bounded claims

What the recorded experiments support

- **Full fine-tuning reaches 84.20%**
  On Food-101, with ImageNet ResNet-50, within the 30-epoch budget.
- **The ranking is stable**
  Full > frozen > scratch in all three seeds.
- **Fine-grained errors remain**
  Confusions concentrate among visually similar dishes.

What we cannot claim yet

- **Not a backbone comparison**
  Only ResNet-50 was evaluated.
- **Not a resolution / augmentation study**
  Those factors were not varied.
- **Not a converged frozen baseline**
  Its validation curves still improve at epoch 30.
- **Baseline gap is confounded**
  Pretraining, input size, LR and parameter count all differ.

APPENDIX

Reproducibility and presentation hand-off

Recorded environment

- 1× NVIDIA A100-SXM4-80GB
- PyTorch 2.11.0+cu128
- CUDA 12.8; torchvision 0.26.0
- Seeds: 42, 67, 69; split seed 42
- Batch 128; 30 epochs; mixed precision

Reproduction path

- Install assignment2/requirements.txt
- Run assignment2/notebooks/main.ipynb
- Dataset downloaded on first use
- Checkpoints: assignment2/checkpoints/
- Logs, curves, predictions, confusion matrices: assignment2/results/

Presentation delivery checklist

- Every member presents or defends a clearly assigned section.
- Do not read slides verbatim; explain the experimental reasoning.
- Use the report/repository for detailed tables, code and full per-class results.
- Publish the video as Public or Unlisted and verify the link.

AI-use disclosure should also be present on the landing page, assignment page, report and AI_USAGE.md as required by the handbook.